ORF 524 — Midterm Exam 1

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Notes:

- Duration of examination: 80 minutes.
  An additional 10 minutes are given to upload your solutions to Gradescope.
- The exam has two questions with five parts each.
  All parts are equally weighted (5 points), for a total of 50 points.
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Gradescope.
- One sheet of personal notes (both sides) is allowed during the examination.
  No other notes, books, electronic devices, AI systems, or outside assistance may be used.
  This is otherwise a closed-book and individual examination.

1 Point Estimation (25 points)

Let $X_1, \ldots, X_n$ be i.i.d. with Lebesgue density

$$f(x;\theta) = \frac{2x}{\theta} \exp\Big(-\frac{x^2}{\theta}\Big)\mathbf{1}\{x > 0\}, \qquad \theta \in \Theta = (0, \infty).$$

You may use without proof that $\mathbb{E}_\theta[X_i] = \sqrt{\pi\theta}/2$ and $\mathbb{V}_\theta[X_i] = \theta(1 - \pi/4)$. Let $\mathsf{Exponential}(\theta)$ denote the distribution with Lebesgue density $\theta^{-1}e^{-y/\theta}\mathbf{1}\{y > 0\}$, which has mean θ and variance θ^2. If $Y_1, \ldots, Y_n$ are i.i.d. $\mathsf{Exponential}(\theta)$, then $\sum_{i=1}^n Y_i \sim \mathsf{Gamma}(n, \theta)$, which has mean $n\theta$ and variance $n\theta^2$. Estimators are compared by their mean squared error, $\mathrm{MSE}_\theta(\widehat{\theta}) = \mathbb{E}_\theta[(\widehat{\theta} - \theta)^2]$. All limits are taken as $n \to \infty$.

1. Show that $T = \sum_{i=1}^n X_i^2$ is a sufficient statistic for θ, and that $X_i^2 \sim \mathsf{Exponential}(\theta)$.

2. Find the maximum likelihood estimator $\widehat{\theta}_{\mathrm{ML}}$ of θ. Show that $\widehat{\theta}_{\mathrm{ML}}$ is unbiased, compute the Fisher information $\mathcal{I}_n(\theta)$ of the sample, and show that $\widehat{\theta}_{\mathrm{ML}}$ is a uniformly minimum variance unbiased (UMVU) estimator of θ.

3. Consider estimators of the form $\widehat{\theta}_{\mathrm{LIN}}(c) = cT$ with $c > 0$. Compute $\mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{LIN}}(c))$, and find the value $c^\star$ that minimizes it. Show that $\mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{LIN}}(c^\star)) < \mathrm{MSE}_\theta(\widehat{\theta}_{\mathrm{ML}})$ for every $\theta \in \Theta$. Is $\widehat{\theta}_{\mathrm{ML}}$ admissible under squared error loss? Explain why your answer does not contradict part 2.

4. Show that $\sqrt{n}(\widehat{\theta}_{\mathrm{ML}} - \theta) \rightsquigarrow \mathcal{N}(0, \theta^2)$, and that $\sqrt{n}(\widehat{\theta}_{\mathrm{LIN}}(c^\star) - \theta)$ has the same limiting distribution. What does this imply about the improvement found in part 3?

5. The method of moments estimator based on $\mathbb{E}_\theta[X_i] = \sqrt{\pi\theta}/2$ is $\widehat{\theta}_{\mathrm{MM}} = 4\overline{X}_n^2/\pi$, where $\overline{X}_n = n^{-1}\sum_{i=1}^n X_i$. Show that $\widehat{\theta}_{\mathrm{MM}}$ is consistent for θ, and find the limiting distribution of $\sqrt{n}(\widehat{\theta}_{\mathrm{MM}} - \theta)$. Which of $\widehat{\theta}_{\mathrm{ML}}$ and $\widehat{\theta}_{\mathrm{MM}}$ would you prefer in large samples, and why?

2 Hypothesis Testing (25 points)

Let $X_1, \ldots, X_n$ be i.i.d. with Lebesgue density

$$f(x;\theta) = \exp\big(-(x-\theta)\big)\mathbf{1}\{x \geq \theta\}, \qquad \theta \in \Theta = \mathbb{R}.$$

You may use without proof that $\mathbb{E}_\theta[X_i] = \theta + 1$ and $\mathbb{V}_\theta[X_i] = 1$. Let $X_{(1)} = \min_{1\leq i\leq n} X_i$ and $\overline{X}_n = n^{-1}\sum_{i=1}^n X_i$, and let $\mathsf{Exponential}(1)$ denote the distribution with Lebesgue density $e^{-y}\mathbf{1}\{y > 0\}$. For fixed $\theta_0 \in \mathbb{R}$ and $\alpha \in (0,1)$, consider the testing problem

$$H_0 : \theta \leq \theta_0 \qquad \text{versus} \qquad H_1 : \theta > \theta_0.$$

Let Φ denote the standard Normal distribution function and $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$. All limits are taken as $n \to \infty$.

1. Show that $X_{(1)}$ is a sufficient statistic for θ, and that $n(X_{(1)} - \theta) \sim \mathsf{Exponential}(1)$ for every $\theta \in \Theta$ and $n \geq 1$.

2. Consider the test that rejects H_0 if and only if $X_{(1)} > \theta_0 + c$, for a constant $c \geq 0$. Find c such that the test has size α.

3. Derive the power function $\beta_n(\theta)$ of the size α test in part 2 for all $\theta \in \Theta$, and show that the test is consistent, that is, $\beta_n(\theta) \to 1$ for every fixed $\theta > \theta_0$. By inverting this family of tests indexed by θ_0, construct a confidence set for θ with exact coverage $1-\alpha$.

4. Let $W_n = \sqrt{n}(\overline{X}_n - 1 - \theta_0)$, and consider instead the test that rejects H_0 if and only if $W_n > z_{1-\alpha}$. Show that this test has limiting size α, that is,

$$\lim_{n\to\infty} \sup_{\theta\leq\theta_0} \mathbb{P}_\theta(W_n > z_{1-\alpha}) = \alpha.$$

5. Consider the local alternatives $\theta_n = \theta_0 + h/\sqrt{n}$ with $h > 0$. Find $\lim_{n\to\infty} \mathbb{P}_{\theta_n}(\text{reject } H_0)$ for the test in part 2 and for the test in part 4. Then repeat both calculations for $\theta_n = \theta_0 + h/n$. Interpret the results in terms of the convergence rates of $X_{(1)}$ and $\overline{X}_n$.